

COURSE CODE : CSA1613

COURSE NAME: Data Warehousing and Data Mining

Q1
A Define Data Mining

Data Mining the process of discovering useful patterns, relationships, and knowledge from large amount of data.

Any Two Basic Data Mining Tasks:-

1. classification 2. clustering

Steps in KDD (Knowledge Discovery in Databases)

1. Data cleaning 2. Data integration 3. Data selection

4. Data Transformation 5. Data Mining

Role of Data Processing

* Removes errors and missing values.

* Improves data quality.

* Increases mining accuracy.

Q1
b Data cleaning

Data cleaning is the process of removing incorrect, incomplete, or duplicate data.

Two Data quality issues.

1. Missing values 2. Noisy data



Handling Missing Values:

* ignore the record

* Fill the mean/median/mode.

* Fill using prediction.

Example: Age = 20, 22, ?, 24 mean = 22

Missing value = 22

Handling Noisy Data

* Binning * Regression * Clustering

Example: 10, 11, 12, 50, 13 Noise = 50 (outlier).

Q₂
a

Data Integration:

Data Integration combines data from multiple source into one dataset.

Two challenges:

1. Data redundancy
2. Data inconsistency.

Handling Redundancy:

* Remove duplicate records * correlation analysis

Handling inconsistency:

* standardize formats * resolve conflicting values.

Covariance Analysis:

Covariance Analysis measure how two variables change together.

Two Covariance Forms:

1. Population covariance

$$\text{cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

2. Sample covariance,

$$\text{cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

3a) Data:

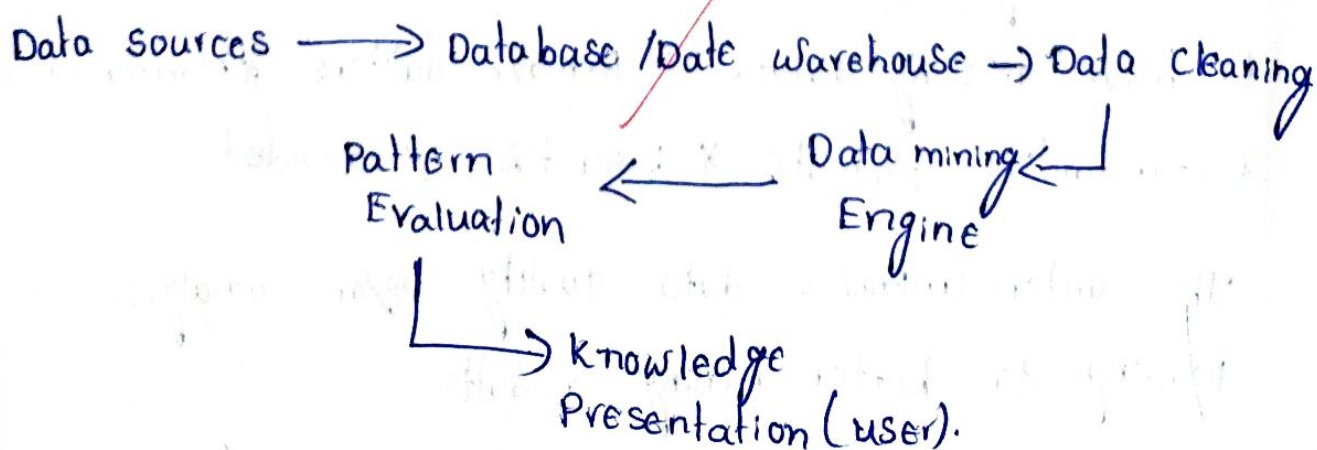
A = 2, 3, 5, 4, 6

B = 5, 8, 10, 11, 14

As A increases, B is also generally increases.

Answer: The two stocks show positive covariance, so, they rise together (not independent).

Q3 b) Architecture of Data Mining system:



Components:-

- * Data sources - collect data.
- * Database / Data warehouse - store data.
- * Data cleaning - Improve quality.
- * Data Mining Engine - Discover patterns.
- * Pattern Evaluation - select useful patterns.
- * Knowledge presentation - Display result.

Q4
a

Given

$$X = \{4, 5, 6, 7, 9\} \quad Y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean of } X = 6.2$$

$$\text{Mean of } Y = 5.8$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = -1.8$$

Sample covariance:

$$\frac{-1.8}{5-1} = -0.45$$

Answer:- covariance = -0.45

Q4
b

Major steps of Data preprocessing:

- * Data cleaning
- * Data integration
- * Data Transformation
- * Data Reduction
- * Data Discretization

Preprocessing pipeline:

- * Remove missing values
- * Remove outliers
- * Normalize data
- * Standardize formats
- * Discretize if needed

This order improves data quality before analysis, leading to better mining results.

Q5) Equal - Frequency (Equi-depth) Binning:

Data sorted:

13, 15, 16, 16, 19, 20, 20, 21, 22

22, 25, 25, 25, 25, 30, 33, 33, 35

35, 35, 35, 36, 40, 45, 46, 52, 70

Total = 27 values

using 3 equal-frequency bins (9 values each)

Bin 1 :- 13, 15, 16, 16, 19, 20, 20, 21, 22

Mean = 18

Bin 2 :- 22, 25, 25, 25, 25, 30, 33, 33, 35

Mean = 28

Bin 3 :- 35, 35, 35, 36, 40, 45, 46, 52, 70

Mean = 44

Smoothed Data :-
18, 18, 18, 18, 18, 18, 18, 18, 18
28, 28, 28, 28, 28, 28, 28, 28, 28
44, 44, 44, 44, 44, 44, 44, 44, 44

Q5) Data Discretization:

Data discretization converts continuous data into intervals (bins).

Importance:

* Reduce data size * Remove noise * Improves mining efficiency.

Equal-width Binning

Equal interval size

Simple

sensitive to outliers

Equal-Frequency Binning

Equal number of values

Handles skewed data better

More balanced bins

Example:-

Data: 10, 20, 30, 40, 50, 60

* Equal width: 10-30, 31-50, 51-70.

* Equal frequency: Each bin contains the same numbers of values.

Impact:-

* Equal-width is easy but affected by outliers.

* Equal-frequency gives balanced groups and often produces better analysis.